VANIER COLLEGE

Advanced Networks (420-521-VA)

Fall 2026

LoRa and Mesh Project

Lab 9 Initial Setup

Topic: LoRa mesh networking

Prepared by

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GitHub repository

https://github.com/Wlszn/LoRaMesh

Date: November 10th, 2026

2. Hardware

Current inventory reported by the team: three ESP32 boards, one per student. Exact models and powered-on status still need confirmation. Complete the device table only after verifying those details.

Type / manufacturer / exact model	Qty.	Status today
LoRa V4 ESP32 LoRa V3 Dev Board Upgraded	2	To be delivered

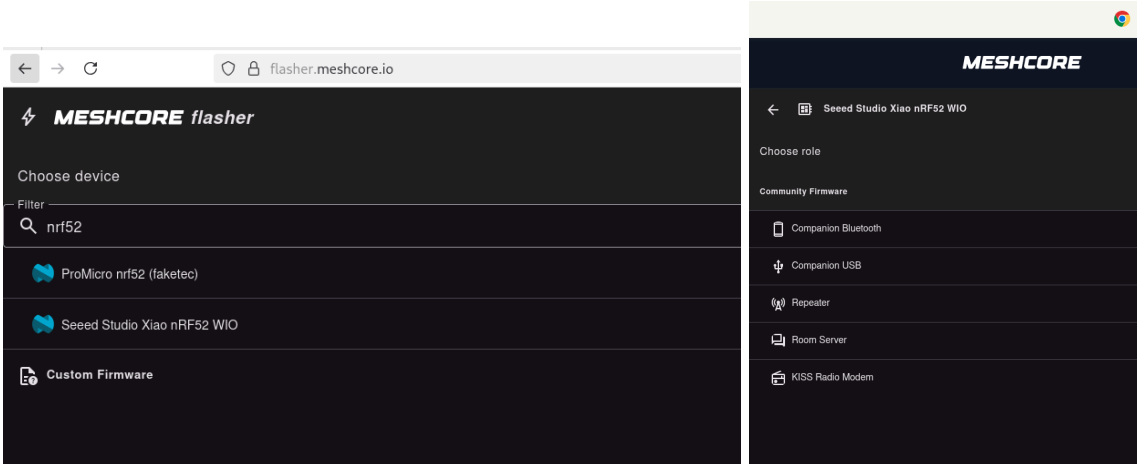
3. Hosting

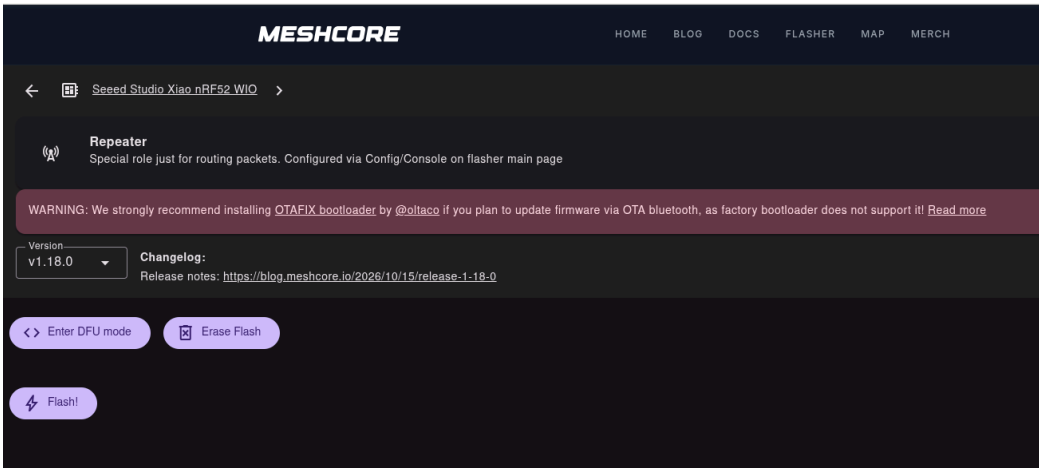
4. Software

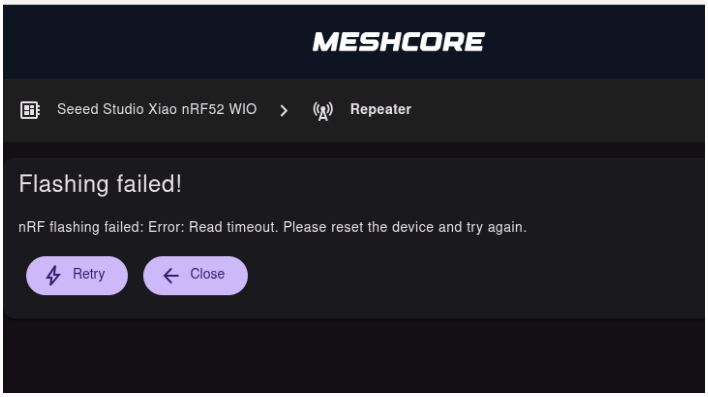
Program	Actual version	Version-check command	Role
[Installed program]	[Observed version]	[Command actually run]	[Project purpose]

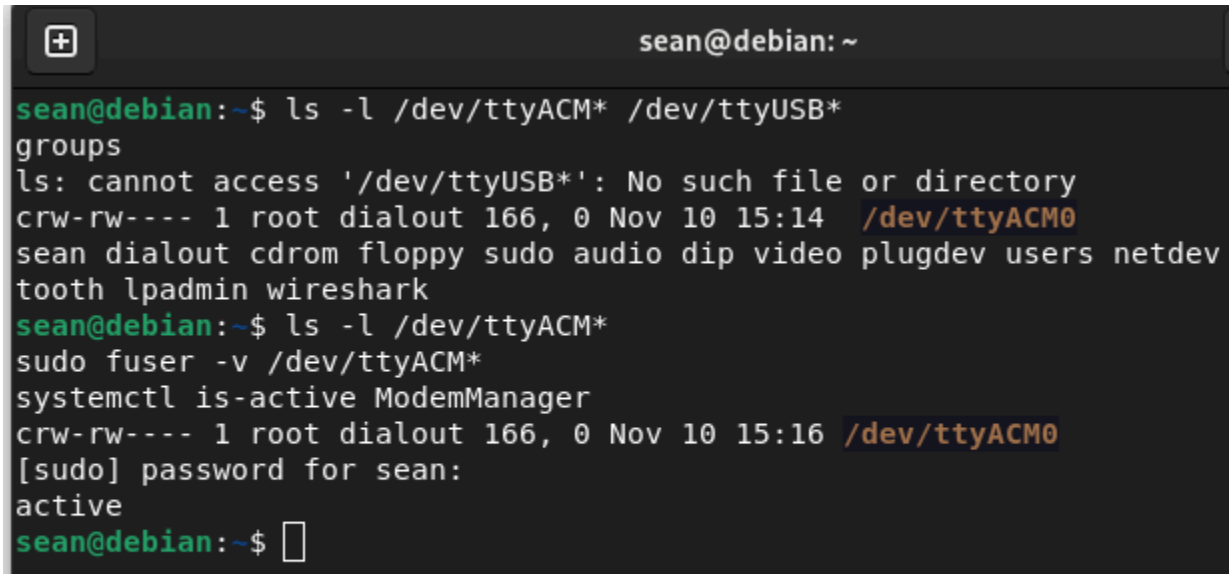
5. What ran today

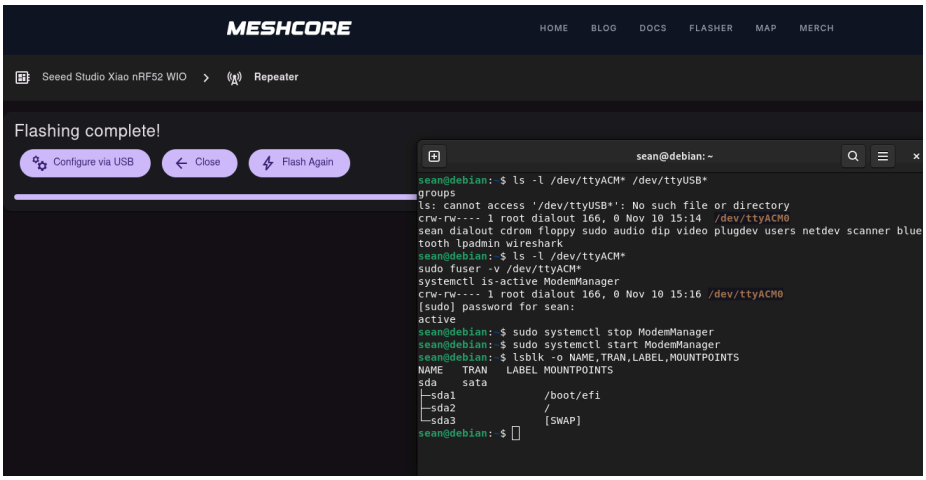
We flashed our Repeater.

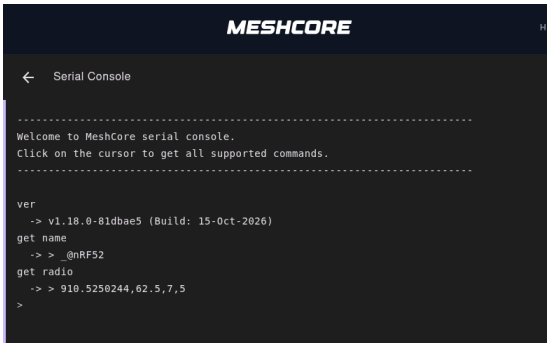



MESHCORE
HOME
BLOG
DOCS
FLASHER
MAP
MERCH
Seeed Studio Xiao nRF52 WIO
Repeater
Special role just for routing packets. Configured via Config/Console on flasher main page
WARNING: We strongly recommend installing OTAFIX bootloader by @oltaco if you plan to update firmware via OTA bluetooth, as factory bootloader does not support it! Read more
Version
v1.18.0
Changelog:
Release notes: https://blog.meshcore.io/2026/10/15/release-1-18-0
Enter DFU mode
Erase Flash
Flash!


MESHCORE
Seeed Studio Xiao nRF52 WIO
Repeater
Flashing failed!
nRF flashing failed: Error: Read timeout. Please reset the device and try again.
Retry
Close


sean@debian: ~
sean@debian:~$ ls -l /dev/ttyACM* /dev/ttyUSB*
groups
ls: cannot access '/dev/ttyUSB*': No such file or directory
crw-rw---- 1 root dialout 166, 0 Nov 10 15:14 /dev/ttyACM0
sean dialout cdrom floppy sudo audio dip video plugdev users netdev
tooth lpadmin wireshark
sean@debian:~$ ls -l /dev/ttyACM*
sudo fuser -v /dev/ttyACM*
systemctl is-active ModemManager
crw-rw---- 1 root dialout 166, 0 Nov 10 15:16 /dev/ttyACM0
[sudo] password for sean:
active
sean@debian:~$


MESHCORE
HOME
BLOG
DOCS
FLASHER
MAP
MERCH
Seeed Studio Xiao nRF52 WIO
Repeater
Flashing complete!
Configure via USB
Close
Flash Again
sean@debian: ~
sean@debian:~$ ls -l /dev/ttyACM* /dev/ttyUSB*
groups
ls: cannot access '/dev/ttyUSB*': No such file or directory
crw-rw---- 1 root dialout 166, 0 Nov 10 15:14 /dev/ttyACM0
sean dialout cdrom floppy sudo audio dip video plugdev users netdev scanner bluetooth lpadmin wireshark
sean@debian:~$ ls -l /dev/ttyACM*
sudo fuser -v /dev/ttyACM*
systemctl is-active ModemManager
crw-rw---- 1 root dialout 166, 0 Nov 10 15:16 /dev/ttyACM0
[sudo] password for sean:
active
sean@debian:~$ sudo systemctl stop ModemManager
sean@debian:~$ sudo systemctl start ModemManager
sean@debian:~$ lsblk -o NAME,TRAN,LABEL,MOUNTPOINTS
NAME TRAN LABEL MOUNTPOINTS
sda sata
├─sda1 /boot/efi
├─sda2 /
└─sda3 [SWAP]
sean@debian:~$


MESHCORE
Serial Console
Welcome to MeshCore serial console.
Click on the cursor to get all supported commands.
ver
-> v1.18.0-81dbae5 (Build: 15-Oct-2026)
get name
-> > _@nRF52
get radio
-> > 910.5250244,62.5,7,5
>

6. Who did what today

Member	Contribution during this lab
Elvéus, Will Kerdy	Set up the Github repo
Hamidouch, Mohamed Amine	Wrote the pdf report
Lussier, Sean	Flashed the Repeater

7. Who does what until the presentation

Responsibilities are not confirmed yet. The task descriptions below are suggestions to discuss with the team.

Task to assign	Responsible member	Target date
Complete repository structure and documentation		
Confirm hardware compatibility and obtain parts		
Install/configure firmware and test communication		
Test repeater, distance, and obstacles		
Complete report, presentation, and evidence		